

Monetary Economics and Policy

Pierpaolo Benigno

Chapter 3: Central Bank Digital
Currency



Prepared by Marcos Consuegra

What We Will Learn

- ▶ How **CBDCs** make central bank liabilities **provide liquidity services**.
- ▶ Why this creates a **spread** between the **policy rate** and the **market rate**.
- ▶ How both the **policy rate** and the **stock of reserves** **affect inflation**.
- ▶ When the model nests the price-level mechanism of Chapter 1.
- ▶ Why **optimal policy** implies supplying enough liquidity to **eliminate the spread**.
- ▶ How the central bank can still **set inflation** through its **policy rate**.
- ▶ Treasury-backed liquidity and **the role of fiscal capacity** in price determination.

Outline

- 1 Introduction
- 2 Model
- 3 Price Determination
- 4 Optimal Liquidity Policy
- 5 Alternative: Treasury-Backed Liquidity
- 6 Key Takeaways

- 1 Introduction
- 2 Model
- 3 Price Determination
- 4 Optimal Liquidity Policy
- 5 Alternative: Treasury-Backed Liquidity
- 6 Key Takeaways

From Crypto to CBDCs: Why This Matters

Context

- ▶ Since Bitcoin's launch in 2009, digital currencies have proliferated, attracting the attention of investors, media, and policymakers.
- ▶ Their novelty does not lie in making payments digital (already possible via cards and bank accounts), but in enabling *decentralized verification* via blockchain, allowing a degree of cash-like anonymity without requiring third-party validation.

Payment System

- ▶ Modern payments are almost fully digital, a significant departure from historical systems where money circulated only in physical form.
- ▶ As Cipolla (1956) observed, in the early Middle Ages “any commodity could serve as a means of exchange, and coins were just one among many”.
- ▶ The dominance of a single, government-issued currency—**monetary sovereignty**—is therefore a relatively recent development.

The Return of Currency Competition

Historical Note

- ▶ Multiple currencies and payment instruments have historically coexisted within the same economy.¹
- ▶ Today, digital assets and global payment platforms are reviving *currency competition*.

Implications for Central Banks (CBs)

- ▶ Competing currencies can erode seigniorage and weaken CBs' control over money's value, especially if settlement ceases to occur in the domestic unit.
- ▶ In response, CBs are exploring **Central Bank Digital Currencies (CBDCs)**.

¹ See also Cipolla (1956, 1982, 1990).

CBDC Form 1: Digital Tokens

(1) CBDC as Digital Tokens replacing Physical Cash

- ▶ Digital tokens represent CB liabilities that store units of account like cash, but their face value could be **digitally adjusted** by the issuer over time.
- ▶ When tokens serve as the store of value, *no-arbitrage* prevents $i_t^X < 0$.
- ▶ However, such tokens can **relax the zero lower bound (ZLB)** by allowing negative interest rates.
- ▶ Still, **price-level control** with tokens alone remains as in Chapters 1–2; novel effects arise when reserves both **pay interest** and **provide liquidity**.

CBDC Form 2: Retail Access to Central Bank Reserves

(2) CBDC as Interest-Bearing, Liquidity-Providing Money

- ▶ In this form, the CBDC is modeled as **interest-bearing deposits** at the CB that are directly accessible to households.
- ▶ These deposits serve as *money-like securities*: they provide **liquidity services** that can enter household utility and also **yield interest**.
- ▶ Introducing these liquidity benefits for interest-bearing reserves alters the mechanism of **price-level determination** discussed in Chapter 2.
- ▶ Whether or not digital tokens circulate is largely irrelevant for this mechanism (under perfect substitutability); **tokens mainly affect the ZLB**.

Outline of the Chapter's Results

Price-Level Control

- ▶ Inflation depends on both the **policy rate** and the **quantity of reserves**, extending the transmission mechanisms discussed in Chapter 1.

Optimal Liquidity Supply

- ▶ The CB can achieve the **optimal supply of liquidity** without affecting the interest-rate policy, price level, or inflation, provided it holds sufficient assets.

Fiscal Coordination

- ▶ Alternatively, optimal liquidity can be achieved through **coordination with the Treasury**, by backing Treasury liabilities, subject to fiscal capacity.

- 1 Introduction
- 2 Model**
- 3 Price Determination
- 4 Optimal Liquidity Policy
- 5 Alternative: Treasury-Backed Liquidity
- 6 Key Takeaways

Model Overview

Economic Setting

- ▶ Infinite-horizon, closed economy under perfect foresight
- ▶ Single consumption good C with constant endowment Y

Agents

- ▶ Representative consumer choosing C_t , bonds, and liquidity balances
- ▶ Treasury issuing debt and levying lump-sum taxes
- ▶ Central Bank (CB) setting the policy rate and managing its balance sheet

CB Liabilities

- ▶ **Digital Tokens** M_t : non-interest-bearing, functioning as digital cash
- ▶ **Reserves** X_t : interest-bearing deposits at the CB
- ▶ Both provide liquidity services through $V\left(\frac{X_t + M_t}{P_t}\right)$

Representative Consumer: Preferences

Intertemporal Utility

$$\sum_{t=t_0}^{\infty} \beta^{t-t_0} \left[U(C_t) + V\left(\frac{X_t + M_t}{P_t}\right) \right] \quad (3.1)$$

Interpretation

- ▶ $\beta \in (0, 1)$: discount factor
- ▶ $U(C_t)$: concave utility in consumption
- ▶ $V(\cdot)$: concave utility in real liquidity services
- ▶ Real liquidity: $q_t = \frac{X_t + M_t}{P_t}$
- ▶ Satiation: $V_q(q_t) = 0$ for $q_t \geq \bar{q}$

Economic Meaning: Reserves and tokens ease transaction frictions by providing liquidity services. Once q_t reaches $\bar{q}$, the household is fully satiated and additional liquidity yields no marginal benefit.

Representative Consumer: Budget Constraint

Budget Constraint at time t

$$B_t + X_t + M_t + P_t C_t + T_t = (1 + i_{t-1})B_{t-1} + (1 + i_{t-1}^X)X_{t-1} + M_{t-1} + P_t Y \quad (3.2)$$

Interpretation:

- ▶ B_t : private/treasury bonds, default-free, unit face value, pay interest i_t
- ▶ X_t : CB reserves, pay interest i_t^X
- ▶ M_t : digital tokens (non-interest-bearing)
- ▶ T_t : lump-sum taxes (if negative, transfers)

Key Features

- ▶ CB's liabilities provide liquidity services.
- ▶ Private or treasury bonds do **not** provide liquidity services $\implies$ their yield i can differ from the policy rate i^X on reserves.

Liquidity Properties and Model Assumptions

Liquidity Properties

- ▶ Both reserves X_t and tokens M_t provide liquidity services through $V(\cdot)$.
- ▶ When the policy rate on reserves is positive ($i_t^X > 0$), reserves strictly dominate tokens as a store of value.
- ▶ Digital tokens could, in principle, be engineered to carry negative returns (e.g. via digital redenomination), thereby relaxing the traditional zero lower bound.

Modeling Assumption

- ▶ For analytical simplicity, we set $M_t = 0$ going forward and treat reserves X_t as the sole liquidity-providing asset. We may also allow the policy rate i_t^X to take negative values if needed.

Borrowing Constraint

Real Borrowing Limit

$$-\frac{(1+i_{t-1})B_{t-1}}{P_t} \leq \frac{(1+i_{t-1}^X)X_{t-1}}{P_t} + \sum_{j=0}^{\infty} R_{t,t+j} \left(Y - \frac{T_{t+j}}{P_{t+j}} \right) < \infty, \quad (3.3)$$

where $R_{t,t+j}$ is the real discount factor from $t+j$ w.r.t. time t .

Real discount factor:

$$R_{t,t+j} = \prod_{i=1}^j \frac{1}{1+r_{t+i-1}}, \quad R_{t,t} \equiv 1, \quad j \geq 1. \quad (3.4)$$

- Note, that the real interest rate r is defined as $1+r_t \equiv (1+i_t)\frac{P_t}{P_{t+1}}$.

Intertemporal Budget Constraint

- An equivalent formulation replaces the borrowing limit (3.3) with:

Intertemporal Budget Constraint:

$$\sum_{t=t_0}^{\infty} R_{t_0,t} \left(C_t + \underbrace{\frac{i_t - i_t^X}{1 + i_t} \frac{X_t}{P_t}}_{\text{liquidity cost}} \right) \leq \frac{(1 + i_{t_0-1})B_{t_0-1} + (1 + i_{t_0-1}^X)X_{t_0-1}}{P_{t_0}} + \sum_{t=t_0}^{\infty} R_{t_0,t} \left(Y - \frac{T_t}{P_t} \right). \quad (3.5)$$

- This extends the Chapter 1 IBC by incorporating the *liquidity cost* of reserves, which captures the opportunity cost of holding liquidity instead of higher-yield bonds.

See Proof in Appendix

Representative Consumer: Problem

- ▶ The representative consumer chooses sequences $\{C_t, B_t, X_t\}_{t=t_0}^{\infty}$ to maximize intertemporal utility at each time $t \geq t_0$:

$$\max_{\{C_t, B_t, X_t\}} \sum_{t=t_0}^{\infty} \beta^{t-t_0} \left\{ U(C_t) + V\left(\frac{X_t}{P_t}\right) \right\}.$$

Subject to:

- ▶ Budget constraint (3.2)
- ▶ Borrowing limit (3.3)
- ▶ Non-negativity constraints $C_t, X_t \geq 0$
- ▶ Given initial values $(1 + i_{t_0-1})B_{t_0-1}, (1 + i_{t_0-1}^X)X_{t_0-1}$

Consumer Optimality

Euler equation

$$\frac{U_c(C_t)}{P_t} = \beta(1 + i_t) \frac{U_c(C_{t+1})}{P_{t+1}}. \quad (3.6)$$

- This optimality condition equates the marginal benefit of consuming today with the marginal benefit of saving.

Intertemporal Budget Constraint

$$\sum_{t=t_0}^{\infty} R_{t_0,t} \left(C_t + \frac{i_t - i_t^X}{1 + i_t} \frac{X_t}{P_t} \right) = \frac{(1 + i_{t_0-1})B_{t_0-1} + (1 + i_{t_0-1}^X)X_{t_0-1}}{P_{t_0}} + \sum_{t=t_0}^{\infty} R_{t_0,t} \left(Y - \frac{T_t}{P_t} \right). \quad (3.7)$$

- The PDV of spending on consumption and holding liquidity equals the initial real asset position plus the PDV of net income.

See Proof in Appendix

Consumer Optimality: Reserves

- ▶ The first-order condition with respect to holding reserves X_t , together with the Euler equation (3.6) implies:

Optimal Demand for Liquidity

$$1 = \underbrace{\frac{V_q\left(\frac{X_t}{P_t}\right)}{U_c(C_t)}}_{\text{non-pecuniary (liquidity) benefit}} + \underbrace{\frac{1 + i_t^X}{1 + i_t}}_{\text{pecuniary return}}.$$

Interpretation:

- ▶ The consumer invests in reserves until the *cost* of one unit equals the sum of:
 - ▶ **Non-pecuniary benefit:** marginal liquidity value $\frac{V_q}{U_c}$,
 - ▶ **Pecuniary benefit:** discounted return $\frac{1 + i_t^X}{1 + i_t}$.

See Proof in Appendix

Liquidity Premium and Spread

- ▶ The **spread in money markets** is therefore given by:

$$\frac{1 + i_t^X}{1 + i_t} = 1 - \varrho_t, \quad \text{where } \varrho_t \equiv \frac{V_q\left(\frac{X_t}{P_t}\right)}{U_c(C_t)}. \quad (3.8)$$

Interpretation:

- ▶ ϱ_t captures the *liquidity premium* from holding reserves, and satisfies $0 \leq \varrho_t < 1$.
- ▶ Because reserves deliver liquidity services, the policy rate i_t^X is typically **lower** than the market rate i_t .
- ▶ Equalization of the two rates ($i_t^X = i_t$) occurs only when liquidity is *satiated*, i.e. when $V_q(\cdot) = 0$.
- ▶ *Economic intuition*: reserves provide non-pecuniary liquidity value, allowing their financial return to lie below the market return without violating arbitrage.

Central Bank: Flow Budget Constraint

- ▶ CB balance sheet:
 - ▶ **Assets:** holdings of short-term private or treasury bonds B_t^C , with interest rate i_t ;
 - ▶ **Liabilities:** central bank reserves X_t^C with interest rate i_t^x ;
 - ▶ **Transfers:** T_t^C to the treasury (if positive).

CB Flow Budget Constraint:

$$B_t^C - X_t^C = (1 + i_{t-1})B_{t-1}^C - (1 + i_{t-1}^x)X_{t-1}^C - T_t^C. \quad (3.9)$$

Treasury: Flow Budget Constraint

- ▶ Treasury debt B_t^F pays the **market nominal interest rate** i_t (no liquidity services).
- ▶ Financing sources:
 - ▶ Lump-sum taxes: T_t
 - ▶ Remittances from the CB: T_t^C
 - ▶ Issuance of new debt: B_t^F

Treasury Flow Budget Constraint:

$$B_t^F = (1 + i_{t-1})B_{t-1}^F - T_t - T_t^C. \quad (3.10)$$

Treasury Solvency Constraint

- ▶ To be considered safe/default-free, Treasury debt and revenues must satisfy:

Solvency Constraint (in real terms):

$$\frac{(1 + i_{t_0-1})B_{t_0-1}^F}{P_{t_0}} = \sum_{t=t_0}^{\infty} R_{t_0,t} \left(\frac{T_t}{P_t} + \frac{T_t^C}{P_t} \right). \quad (3.11)$$

- ▶ Taxes should adjust to repay debt with certainty at any equilibrium path of prices and CB remittances.
- ▶ The imposed equality, rather than $\leq$ ensures that the Treasury does not levy more taxes than necessary (as discussed in Chapter 1).

Market Clearing Conditions

- ▶ Debt issued by the Treasury is held either by the CB or the consumer. CB reserves are held only by the consumer.

Asset Market Clearing:

$$B_t^F = B_t^C + B_t, \quad (3.12)$$

$$X_t^C = X_t. \quad (3.13)$$

Goods Market Clearing:

$$C_t = Y.$$

- ▶ Consumption equals output, consistent with a closed economy with no investment or government purchases beyond transfers.

Equilibrium Condition: Fisher Equation

- ▶ Substituting $C_t = Y$ into the Euler equation (3.6) implies the Fisher equation:

Fisher Equation

$$1 + i_t = \frac{1}{\beta} \frac{P_{t+1}}{P_t} \quad (3.14)$$

- ▶ Links the nominal market interest rate to inflation.

Equilibrium Condition: Liquidity Market

- Using $C_t = Y$, setting $U_c(Y) = 1$, and inserting in (3.8), gives:

Liquidity Market in Equilibrium

$$\frac{1 + i_t^X}{1 + i_t} = 1 - V_q \left(\frac{X_t}{P_t} \right). \quad (3.15)$$

- Spread in money markets $(i_t - i_t^X)$ depends on the liquidity premium.

Equilibrium Condition: Intertemporal Budget Constraint

- Using the goods and asset market clearing in the intertemporal budget constraint (3.7):

Intertemporal Resource Constraint in Equilibrium

$$\sum_{t=t_0}^{\infty} \beta^{t-t_0} \left(\frac{T_t}{P_t} + \frac{i_t - i_t^X}{1 + i_t} \frac{X_t}{P_t} \right) = \frac{(1 + i_{t_0-1})B_{t_0-1} + (1 + i_{t_0-1}^X)X_{t_0-1}}{P_{t_0}}, \quad (3.16)$$

or, equivalently, using the treasury solvency condition (3.11):

$$\sum_{t=t_0}^{\infty} \beta^{t-t_0} \left(\frac{T_t^C}{P_t} \right) = \frac{(1 + i_{t_0-1})B_{t_0-1}^C - (1 + i_{t_0-1}^X)X_{t_0-1}}{P_{t_0}} + \sum_{t=t_0}^{\infty} \beta^{t-t_0} \left(\frac{i_t - i_t^X}{1 + i_t} \frac{X_t}{P_t} \right). \quad (3.17)$$

Equilibrium Definition

Equilibrium under CBDC: A set of sequences

$$\{P_t, i_t^X, i_t, B_t^C, X_t, T_t^C\}_{t=t_0}^{\infty},$$

with $\{P_t, X_t, B_t^C, T_t^C\}_{t=t_0}^{\infty} \geq 0$, satisfying:

- ▶ CB Flow Budget Constraint (3.9)
- ▶ Euler equation (3.14)
- ▶ Liquidity market equilibrium (3.15)
- ▶ Intertemporal resource constraint (3.17)

with $i_t \geq i_t^X$, given initial conditions $(1 + i_{t_0-1})B_{t_0-1}^C, (1 + i_{t_0-1}^X)X_{t_0-1}$.

- ▶ The CB sets policy in terms of the sequences $\{i_t^X, X_t, T_t^C\}_{t=t_0}^{\infty}$.

- 1 Introduction
- 2 Model
- 3 Price Determination**
- 4 Optimal Liquidity Policy
- 5 Alternative: Treasury-Backed Liquidity
- 6 Key Takeaways

Price Determination under CBDC

- ▶ Determination of price level is more complex than in Chapters 1–2.
- ▶ The relevant interest rate in the Fisher equation (3.14) is the **market rate**, **not** the policy rate. Market and policy rates are linked through the liquidity premium.
- ▶ When reserves exceed the full satiation level, market and policy rates coincide, nesting the setup of Chapter 1.

Initial Price Level: Remittances Policy

Policy Setup

- ▶ The CB sets a constant policy rate

$$i_t^X = \frac{1}{\beta} - 1,$$

and chooses a reserve path $\{X_t\}_{t=t_0}^{\infty}$.

CB Remittances Rule

$$\frac{T_t^C}{P_t} = (1 - \beta)\tau^C + (i_{t-1} - i_{t-1}^X) \frac{X_{t-1}}{P_t}. \quad (3.18)$$

- ▶ This specification allows the CB to rebate the implicit seigniorage generated when $i_t > i_t^X$.
- ▶ Plugging (3.18) into the CB intertemporal constraint (3.17) determines the initial price level.

Initial Price Level: Remittances Policy

Initial Price Level

$$P_{t_0} = \frac{(1 + i_{t_0-1})(B_{t_0-1}^C - X_{t_0-1})}{\tau^C} \quad (3.19)$$

Interpretation

- ▶ A positive and determinate price level requires numerator and denominator to have the same sign:

$$\text{e.g. } \tau^C > 0 \quad \text{and} \quad B_{t_0-1}^C - X_{t_0-1} > 0.$$

- ▶ When seigniorage revenues arise ($i_t > i_t^X$), the term τ^C may be *negative* while still generating non-negative remittances.
- ▶ Thus, $P_{t_0} > 0$ is compatible with **negative CB net worth**:

$$B_{t_0-1}^C - X_{t_0-1} < 0,$$

provided $\tau^C < 0$.

Equilibrium Price Dynamic

- Equilibrium price sequence is determined by combining the Fisher equation (3.14) with the equilibrium in the liquidity market (3.15):

$$\frac{P_{t+1}}{P_t} = \beta \frac{1 + i^X}{1 - V_q(X_t/P_t)} \quad (3.20)$$

- Starting from P_{t_0} as determined by (3.19), and given i^X and X_{t_0} , equation (3.20) recursively determines P_t for all $t > t_0$.
- **Novelty:** Inflation additionally depends on the supply of CB reserves.

Reserve Supply: Mechanism of Adjustment

Experiment: increase in reserves X_{t_0} , with $i^X = 1/\beta - 1$ and τ^C fixed.

- ▶ A higher X_{t_0} reduces the liquidity premium via equation (3.20), lowering the market interest rate i_{t_0} .
- ▶ The reserve expansion is accompanied by a proportional increase in the CB's bond holdings B_t^C .
- ▶ This follows from combining the remittances policy (3.18) with the CB flow budget constraint (3.9):

$$B_t^C - X_t = (1 + i_{t-1})(B_{t-1}^C - X_{t-1}) - (1 - \beta)P_t\tau^C. \quad (3.21)$$

- ▶ Over time, the lower market interest rate i_t reduces inflation through equation (3.20).
- ▶ As prices fall gradually after t_0 , they stabilize once the liquidity premium vanishes (i.e., when reserves reach the satiation level, $V_q \rightarrow 0$).

Interpreting the Effect of Reserve Expansion

- ▶ It may seem counterintuitive that a higher level of CB reserves lowers the price level.
- ▶ This result arises under **flexible prices**, where the **Fisher relation** (3.14) holds: an increase in reserves lowers the liquidity premium, reducing the market interest rate, which in turn lowers inflation.
- ▶ Under **price rigidity**, however, the Fisher relation no longer holds. In that case, a lower market interest rate stimulates aggregate demand, output, and inflation, as discussed in Chapter 8.

- 1 Introduction
- 2 Model
- 3 Price Determination
- 4 Optimal Liquidity Policy**
- 5 Alternative: Treasury-Backed Liquidity
- 6 Key Takeaways

Optimal Liquidity Policy: Overview

- ▶ Consider the liquidity policy that maximizes utility (3.1), with $M_t = 0$ and $C_t = Y$.
- ▶ Optimal liquidity policy requires **satiation of liquidity**, i.e. $V_q(\cdot) = 0$, which implies

$$i_t = i_t^X \quad (\text{zero spread}).$$

- ▶ Unlike in Chapter 2, where $i_t^X = 0$, satiation does *not* imply a deflation rate of β .
- ▶ With $V_q(\cdot) = 0$, the CB can set inflation at any desired rate directly by choosing i_t^X in the Fisher relation (3.14).
- ▶ Remittances and balance sheet policies remain essential for **price-level determinacy**, exactly as in Chapter 1.

Liquidity Satiation and Price-Level Determinacy

- ▶ Maintaining the remittances policy (3.18), when $i_t = i_t^X$, requires $\tau^C > 0$ so that the CB does not require support from the treasury.
- ▶ Price-level determinacy through (3.19) in turn requires that **CB assets exceed its liabilities**, ensuring positive net worth.
- ▶ Combining (3.19) with the CB flow constraint (3.21) yields:

$$\frac{X_t}{P_t} = \frac{B_t^C}{P_t} - \beta \tau^C.$$

- ▶ **Interpretation:** To maintain liquidity satiation, $X_t/P_t \geq \bar{q}$, the CB must hold a sufficient stock of *default-free assets* in excess of its liabilities $\implies$ is it not just about liquidity supply.

Friedman (1960), Narrow Banking, and CBDC

- ▶ Friedman (1960) advocated a monetary system with:
 - ▶ interest paid on reserves, and
 - ▶ **100% reserve banking** (narrow banking).
- ▶ A framework that closely resembles a retail CBDC environment.
- ▶ In such a system:
 - ▶ the CB supplies all liquidity at a policy rate $i^X > 0$,
 - ▶ holds a portfolio of *default-free* assets,
 - ▶ and may impose reserve requirements on intermediaries that issue liquid liabilities to the public.
- ▶ **Bottom line:** With a sufficiently large stock of safe assets, the CB can simultaneously:
 - ▶ *control the price level* through its remittances and balance-sheet policy, and
 - ▶ *optimally supply liquidity* through (X_t, i^X) .

- 1 Introduction
- 2 Model
- 3 Price Determination
- 4 Optimal Liquidity Policy
- 5 Alternative: Treasury-Backed Liquidity**
- 6 Key Takeaways

Extension: Treasury Debt Supplying Liquidity Services

- ▶ Assume that the CB *backs* Treasury debt (as discussed in Chapter 1), so that Treasury solvency constraint does not apply. Moreover assume that Treasury securities held by households provide the *same* liquidity services as reserves.
- ▶ Real liquidity becomes

$$q_t = \frac{B_t + X_t}{P_t},$$

and enters the liquidity-utility function $V(\cdot)$ in (3.1).

- ▶ With CB backing, the Treasury solvency constraint (3.11) no longer applies, since Treasury liabilities inherit the safety of central bank liabilities.
- ▶ The key equation determining the price level is now the consumer intertemporal condition (3.16), which consolidates monetary and fiscal components.

Policy specification

- **Policy:** Consider, for simplicity, an interest-rate peg $i_t^X = i^X$ for each $t \geq t_0$ and the following specification of the real tax policy:

$$\frac{T_t}{P_t} = (1 - \beta)\tau - (i_{t-1} - i_{t-1}^X) \frac{B_{t-1} + X_{t-1}}{P_t}. \quad (3.22)$$

- **Interpretation:** The tax rule has a constant real component, τ , and a second component that transfers seigniorage revenues (i.e., $i - i^X$) to consumers.

Price Level Determination

- Substituting the tax rule (3.22) into the critical equation (3.16) yields:

$$\tau = \frac{(1 + i_{t_0-1})(B_{t_0-1} + X_{t_0-1})}{P_{t_0}},$$

and therefore

$$P_{t_0} = \frac{(1 + i_{t_0-1})(B_{t_0-1} + X_{t_0-1})}{\tau}.$$

- **Interpretation:** The price level P_{t_0} is determined as proportional to the total initial liabilities of the government and inversely proportional to τ .

Fiscal Capacity and Real Liquidity Supply

- ▶ We can also use the consolidated budget constraint of the government to find that at each point in time the real value of total government liabilities is constant:

$$\frac{B_t + X_t}{P_t} = \beta\tau. \quad (3.23)$$

- ▶ **Interpretation:** The parameter τ represents the present discounted value of real taxes. It can be interpreted as the government's **fiscal capacity** that backs the real value of its net liabilities.

See Proof in Appendix

Inflation Determination

- The optimality condition (3.15) still holds, but $V_q(\cdot)$ now takes total real liabilities as its argument. Combining with the Fisher equation (3.14):

$$\frac{P_{t+1}}{P_t} = \beta \frac{(1 + i^X)}{1 - V_q\left(\frac{B_t + X_t}{P_t}\right)},$$

- Substituting in our result from (3.23):

$$\frac{P_{t+1}}{P_t} = \beta \frac{(1 + i^X)}{1 - V_q(\beta\tau)}. \quad (3.24)$$

- **Interpretation:** The inflation rate is determined by both the monetary policy peg i^X and the fiscal capacity τ , which sets the level of real liquidity.

Implications: Joint Monetary–Fiscal Control

- ▶ Both monetary and fiscal policies play pivotal roles in determining the inflation rate.
- ▶ **Monetary policy** directly influences inflation through the interest rate on reserves (i^X).
- ▶ **Fiscal policy** shapes inflation by governing the overall liquidity supply ($\beta\tau$) and, consequently, the liquidity premium $V_q(\beta\tau)$.
- ▶ To achieve the optimal supply of liquidity, the treasury must be able to appropriately set τ .
- ▶ **Implication:** The fiscal capacity (τ) is crucial for determining the price level and real liquidity. Price and inflation determination are a **shared concern** of both monetary and fiscal policies, resting on the CB's support for the treasury.

- 1 Introduction
- 2 Model
- 3 Price Determination
- 4 Optimal Liquidity Policy
- 5 Alternative: Treasury-Backed Liquidity
- 6 Key Takeaways**

Key Takeaways

- ▶ Liquidity-providing CBDC deposits introduce a **spread** between **the policy rate** and **the market rate**.
- ▶ The **quantity of reserves affects inflation** via the liquidity premium; price-level determinacy still needs remittances/tax anchors.
- ▶ **Optimal policy satiates liquidity**, pays interest on reserves, and maintains enough safe assets.
- ▶ Under CB-backed Treasury debt, inflation is **co-determined by the policy rate and fiscal capacity**.

7 Appendix: Derivations

8 Bibliography

Proof: Borrowing Constraint (3.5)

The derivation follows the same steps as in the Appendix of Chapter 1. Just note that budget constraint (3.2) can be written by dividing it for P_t and setting $M_t = M_{t-1} = 0$ as

$$\frac{B_t}{P_t} + \frac{X_t}{P_t} + C_t + \frac{T_t}{P_t} = \frac{(1 + i_{t-1})B_{t-1} + (1 + i_{t-1}^X)X_{t-1}}{P_t} + Y.$$

Denote

$$\mathcal{W}_{t-1} = (1 + i_{t-1})B_{t-1} + (1 + i_{t-1}^X)X_{t-1}$$

and observe that

$$B_t = \frac{\mathcal{W}_t}{1 + i_t} - \frac{1 + i_t^X}{1 + i_t} X_t.$$

It follows that

$$\frac{\mathcal{W}_t}{P_t(1+i_t)} + \frac{(i_t - i_t^X)}{(1+i_t)} \frac{X_t}{P_t} + C_t + \frac{T_t}{P_t} = \frac{\mathcal{W}_{t-1}}{P_t} + Y.$$

Starting from the above equation, equation (3.5) can be derived following similar steps as in the Appendix of Chapter 1.

[Back to Intertemporal Budget Constraint](#)

Proof: Euler Equation

A representative household maximizes lifetime utility:

$$\max_{\{C_t, B_t, X_t\}_{t=t_0}^{\infty}} \sum_{t=t_0}^{\infty} \beta^{t-t_0} \left\{ U(C_t) + V\left(\frac{X_t}{P_t}\right) \right\} \quad (3.25)$$

subject to the budget constraint

$$B_t + X_t + P_t C_t + T_t = (1 + i_{t-1})B_{t-1} + (1 + i_{t-1}^X)X_{t-1} + P_t Y \quad (3.26)$$

and the borrowing limit

$$-\frac{(1 + i_{t-1})B_{t-1}}{P_t} \leq \frac{(1 + i_{t-1}^X)X_{t-1}}{P_t} + \sum_{j=0}^{\infty} R_{t,t+j} \left(Y - \frac{T_{t+j}}{P_{t+j}} \right) \quad (3.27)$$

Lagrangian

Defining the Lagrangian with multipliers λ_t for the budget constraint (3.26) and $\tilde{\zeta}_t$ for the borrowing limit (3.27), we get the following Lagrangian:

$$\begin{aligned} \mathcal{L} = \sum_{t=t_0}^{\infty} \beta^{t-t_0} & \left[U(C_t) + V\left(\frac{X_t}{P_t}\right) \right. \\ & + \lambda_t \left((1 + i_{t-1})B_{t-1} + (1 + i_{t-1}^X)X_{t-1} + P_t Y - B_t - X_t - P_t C_t - T_t \right) \\ & \left. + \tilde{\zeta}_t \left((1 + i_{t-1})B_{t-1} + (1 + i_{t-1}^X)X_{t-1} + P_t \sum_{j=0}^{\infty} R_{t,t+j} \left(Y - \frac{T_{t+j}}{P_{t+j}} \right) \right) \right]. \end{aligned} \quad (3.28)$$

In an interior optimum, we can set $\tilde{\zeta}_s = 0 \forall s$.

First Order Conditions and Euler Equation

The first-order conditions (FOCs) are:

$$\frac{\partial \mathcal{L}}{\partial C_t} : \quad \beta^{t-t_0} U_c(C_t) - \beta^{t-t_0} \lambda_t P_t = 0 \quad \Rightarrow \quad \lambda_t = \frac{U_c(C_t)}{P_t}, \quad (3.29)$$

$$\frac{\partial \mathcal{L}}{\partial B_t} : \quad -\beta^{t-t_0} \lambda_t + \beta^{t+1-t_0} \lambda_{t+1} (1 + i_t) = 0, \quad (3.30)$$

$$\frac{\partial \mathcal{L}}{\partial X_t} : \quad \beta^{t-t_0} \frac{1}{P_t} V_q \left(\frac{X_t}{P_t} \right) - \beta^{t-t_0} \lambda_t + \beta^{t+1-t_0} \lambda_{t+1} (1 + i_t^X) = 0. \quad (3.31)$$

Dividing by β^{t-t_0} the FOC for B_t (3.30) reduces to

$$-\frac{\lambda_t}{1+i_t} + \beta\lambda_{t+1} = 0 \quad \Rightarrow \quad \lambda_t = \beta(1+i_t)\lambda_{t+1}. \quad (3.32)$$

Substituting $\lambda_t = \frac{U_c(C_t)}{P_t}$ and rearranging:

$$U_c(C_t) = \beta(1+i_t)\frac{P_t}{P_{t+1}}U_c(C_{t+1}) \quad (3.33)$$

Or using the (gross) real rate:

$$U_c(C_t) = \beta(1+r_t)U_c(C_{t+1}) \quad (3.34)$$

[Back to Euler Equation](#)

Consumer Optimality: Proof Reserves

Dividing the FOC (3.31) by β^{t-t_0} , we obtain:

$$\frac{1}{P_t} V_q \left(\frac{X_t}{P_t} \right) - \lambda_t + \beta \lambda_{t+1} (1 + i_t^X) = 0 \quad (3.35)$$

Substituting $\lambda_t = \frac{U_c(C_t)}{P_t}$ and rearranging:

$$\frac{U_c(C_t)}{P_t} = \frac{1}{P_t} V_q \left(\frac{X_t}{P_t} \right) + \beta \frac{U_c(C_{t+1})}{P_{t+1}} (1 + i_t^X) \quad (3.36)$$

Dividing by $\frac{U_c(C_t)}{P_t}$ and using (3.33):

$$1 = \frac{V_q \left(\frac{X_t}{P_t} \right)}{U_c(C_t)} + \frac{1 + i_t^X}{1 + i_t}. \quad (3.37)$$

[Back to Liquidity Optimality](#)

Fiscal Capacity Derivation

Combining the government budget constraints, (3.9) and (3.10), while using the asset market clearings, we obtain:

$$B_t + X_t = (1 + i_{t-1})(B_{t-1} + X_{t-1}) - T_t \quad (3.38)$$

Dividing by P_t :

$$\frac{B_t + X_t}{P_t} = \frac{(1 + i_{t-1})(B_{t-1} + X_{t-1})}{P_t} - \frac{T_t}{P_t} \quad (3.39)$$

Inserting the real tax policy (3.22) gives :

$$\frac{B_t + X_t}{P_t} = \beta\tau \quad (3.40)$$

[Back to Fiscal Capacity](#)

7 Appendix: Derivations

8 Bibliography

References

-  Cipolla, Carlo Maria (1956). *Money, Prices, and Civilization in the Mediterranean World*. Princeton: Princeton University Press.
-  — (1982). *The Monetary Policy of Fourteenth-Century Florence*. Berkeley: University of California Press.
-  — (1990). *Il governo della moneta a Firenze e a Milano nei secoli XIV–XVI*. Bologna: Il Mulino.
-  Friedman, Milton (1960). *A Program for Monetary Stability*. New York: Fordham University Press.